OOAD Project — Realtime Delivery System

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Team Details

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Problem Statement: Realtime Delivery System

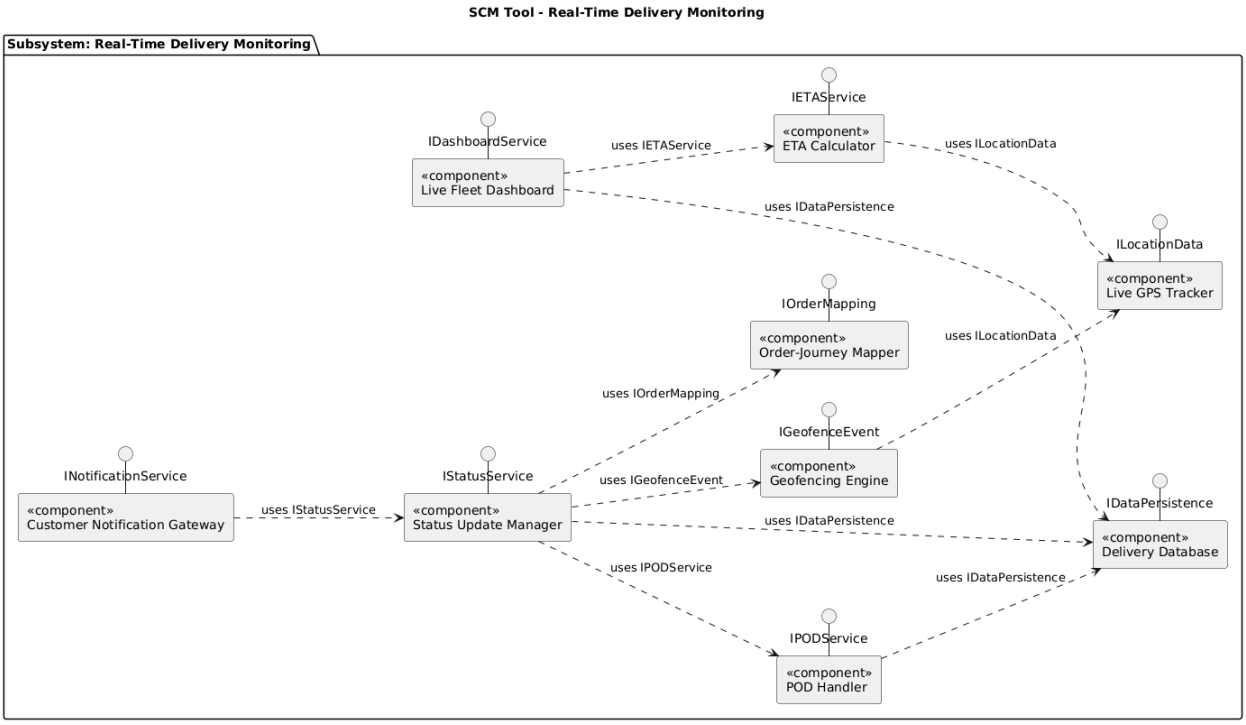
List of Components

The Real-Time Delivery Monitoring subsystem consists of the following components:

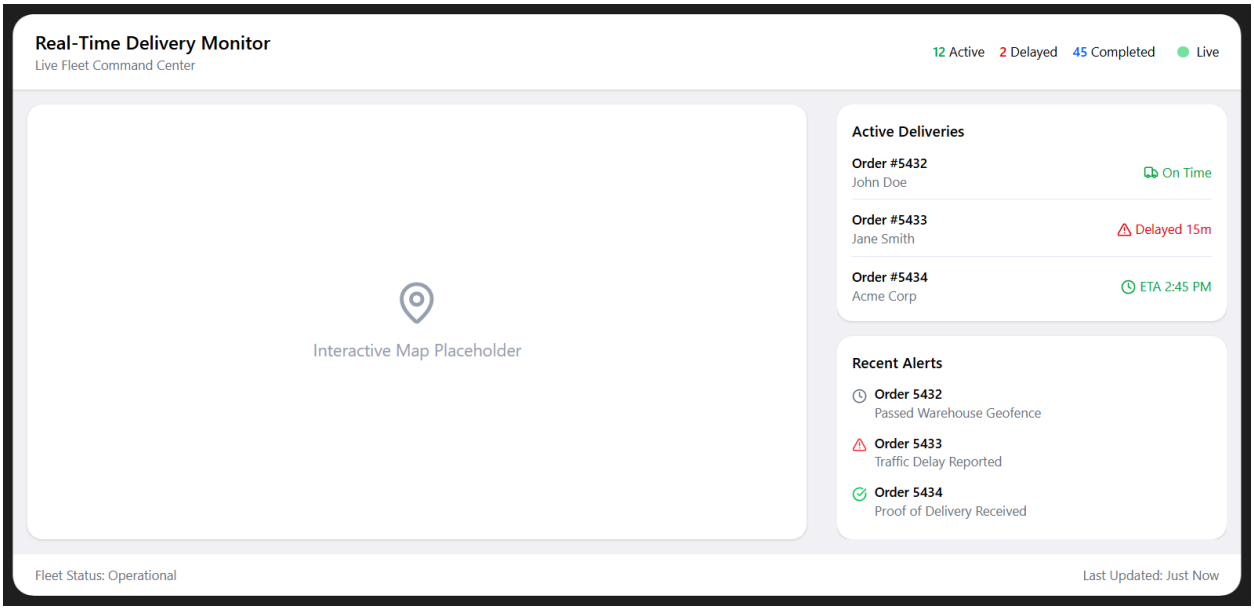
1. **Live GPS Tracker**
2. **Geofencing Engine**
3. **Order-Journey Mapper**
4. **ETA Calculator**
5. **Status Update Manager**
6. **Proof of Delivery (POD) Handler**
7. **Customer Notification Gateway**
8. **Live Fleet Dashboard**
9. **Delivery Database**

#	Component Name	Inputs	Outputs
1	Live GPS Tracker	Device ID, Latitude, Longitude, Timestamp	Structured Location Data
2	Geofencing Engine	GPS Data, Geo-fence Coordinates	Entry/Exit Trigger Events
3	Order-Journey Mapper	Order ID, Rider ID, Route Plan	Order-Vehicle Mapping
4	ETA Calculator	Current Location, Destination, Traffic Data	Estimated Time of Arrival
5	Status Update Manager	Geofence Trigger, Driver Input	Updated Delivery Status
6	POD Handler	Signature, Image, Notes	Delivery Confirmation
7	Customer Notification Gateway	Order ID, Status Message	SMS/Email Notification
8	Live Fleet Dashboard	System Data	Visual Map & Delivery Overview
9	Delivery Database	System Updates	Persistent Data Storage

Component Diagram


SCM Tool - Real-Time Delivery Monitoring
Subsystem: Real-Time Delivery Monitoring
IETAService
«component»
ETA Calculator
IDashboardService
«component»
Live Fleet Dashboard
uses IETAService
uses ILocationData
uses IDataPersistence
ILocationData
«component»
Live GPS Tracker
IOrderMapping
«component»
Order-Journey Mapper
uses ILocationData
uses IOrderMapping
IGeofenceEvent
«component»
Geofencing Engine
INotificationService
«component»
Customer Notification Gateway
uses IStatusService
IStatusService
«component»
Status Update Manager
uses IGeofenceEvent
uses IDataPersistence
IDataPersistence
«component»
Delivery Database
uses IPODService
uses IDataPersistence
IPODService
«component»
POD Handler

Wire Frame


Real-Time Delivery Monitor
Live Fleet Command Center
12 Active
2 Delayed
45 Completed
Live
Interactive Map Placeholder
Active Deliveries
Order #5432
John Doe
On Time
Order #5433
Jane Smith
Delayed 15m
Order #5434
Acme Corp
ETA 2:45 PM
Recent Alerts
Order 5432
Passed Warehouse Geofence
Order 5433
Traffic Delay Reported
Order 5434
Proof of Delivery Received
Fleet Status: Operational
Last Updated: Just Now